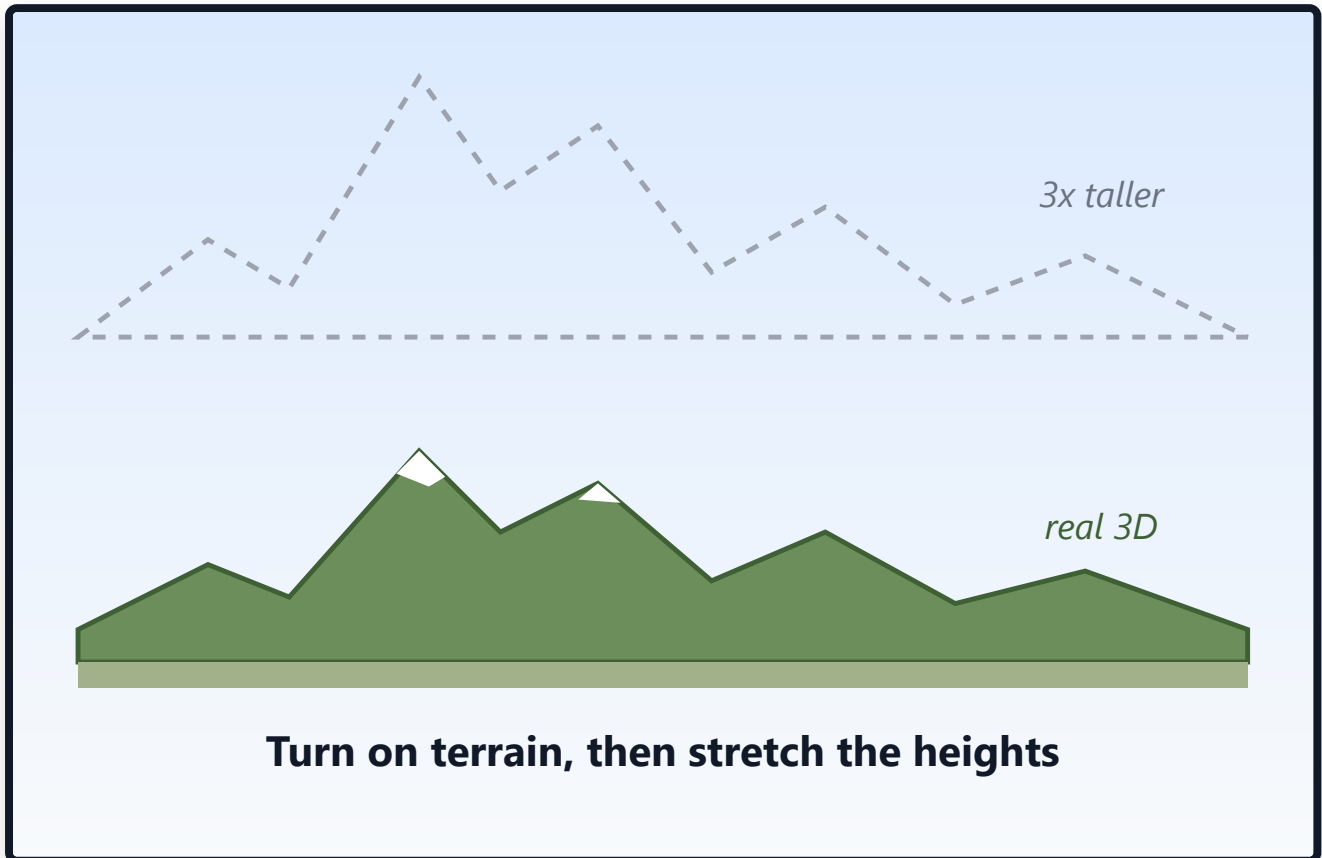


Himalayas in 3D

Fly over the highest mountains on Earth using GeoLibre



Same mountains, two ways: real height, and height stretched three times.

What you'll use

- **GeoLibre**: the same free, web-based GIS tool from the earlier exercises
- **india_boundary.geojson**: the outline of India, from Exercise 1
- No new data! GeoLibre loads the mountain heights by itself.

The Earth Is Not Flat (Even on a Map)

Every map you've made so far has been flat, like a sheet of paper. But the real Earth has mountains, valleys, and plains. Today you'll switch on **3D terrain** and see India's ground rise up off the screen.

We'll head to the **Himalayas**, the highest mountains on Earth, and find out just how high they really are. Then you'll compare them with the place where *you* live.

GIS WORD OF THE DAY

Elevation: how high a place is above the level of the sea, usually measured in metres. The beach is close to 0 m. The top of Mount Everest is 8,849 m.

FUN FACT

The heights GeoLibre shows were measured from space! A space shuttle and satellites bounced radar signals off the ground to map the height of almost the whole planet. That's remote sensing, just like in the intro.

Here's the plan for today, in seven steps:

1. Set up a satellite map
2. Turn on 3D terrain
3. Tilt and turn the map
4. Stretch the mountains
5. Find out how high things are
6. Measure up and down
7. Save your best view

1 Set Up a Satellite Map

Mountains look best in a real photo from space.

1. Open GeoLibre and click **Launch GeoLibre Web**.
2. Switch the basemap to **Google Satellite**, just like in Exercise 1.
3. Add **india_boundary.geojson** with a transparent fill and a bright outline, like yellow, so it shows up against the mountains.
4. Zoom in on the top of India. Look for the white, wrinkled band of snow: those are the Himalayas.

TIP

From high above, the Himalayas look flat, like a painting. Keep that in mind; it's about to change!

2

Turn On 3D Terrain

Now let's give the map real height.

1. Open the **Controls** menu and turn on **Terrain**.
2. Look at the map. Did anything change? Probably not much yet! When you look straight down from above, mountains still look flat.
3. That's because we need to look from the side. That's the next step.

GIS WORD OF THE DAY

Terrain: the shape of the land: its hills, mountains, valleys, and plains. A GIS stores terrain as a grid of elevation numbers, one for each small square of ground. This kind of grid is called a **DEM** (Digital Elevation Model).

FUN FACT

A DEM is a type of **raster** data, the grid-of-squares kind you read about in Exercise 1. Instead of colours, each square holds a height.

Time to stop looking down like a bird and start looking across like a pilot.

1. **Tilt:** hold **Ctrl** and drag the map up or down with your mouse (or press **Shift** + ↑ / ↓). The mountains suddenly stand up!
2. **Turn:** hold the **right mouse button** and drag left or right (or press **Shift** + ← / →) to spin around the mountains.
3. Zoom in on a big peak and walk around it by turning slowly.
4. Lost? Click the **compass** to point north again.

GIS WORD OF THE DAY

Perspective view: looking at the map at an angle, like from an aeroplane window, instead of straight down. Things far away look smaller, just like in real life.

TIP

Try finding where the mountains suddenly stop and the flat plains begin. That sharp edge runs right across the north of India.

4

Stretch the Mountains

Even the Himalayas can look a bit small from far away. GIS can make them taller, just for looking.

1. **Double-click** the **Terrain** control to open its setting.
2. Set the vertical exaggeration to **2**. Every mountain is now drawn twice as tall.
3. Try **3**, then **5**. How do the mountains look now?
4. Look at a flat area, like the plains near the Ganga river. Does stretching change it much? Why not?
5. Set it back to **1** to see the true shape again.

GIS WORD OF THE DAY

Vertical exaggeration: stretching heights to make the shape of the land easier to see. It's useful, but remember: it isn't how the land really looks!

FUN FACT

Stretching flat land does almost nothing. Two times zero is still zero! That's why the plains stay flat while the mountains shoot up.

Let's read the actual heights from the map.

1. Open the **Controls** menu and turn on **Elevation**.
2. Move your mouse over the map. The height of the ground under your pointer appears in the **status bar** at the bottom of the screen.
3. Find each place below, read its height, and write it in the table. Then add two places of your own choice.

Place	Your reading	Real height
Kangchenjunga (Sikkim), India's highest peak		8,586 m
Nanda Devi (Uttarakhand)		7,816 m
Mount Everest (Nepal)		8,849 m
Varanasi, on the Ganga plains		about 80 m
Bangalore, on the Deccan plateau		about 920 m
Your choice: _____		
Your choice: _____		

TIP

Your readings may be a bit lower than the real heights for the sharp peaks. The DEM squares are wide, so each one shows an average height, and the very tip of a peak gets smoothed out.

6

Measure Up and Down

In Exercise 2 you measured distances on a flat map. With terrain on, GeoLibre can measure along the ups and downs of the ground too.

1. Make sure **Terrain** is still on.
2. Open **Controls** → **Measure** and choose **Distance**.
3. Click **Start**, then click a path across a mountain range: from a valley, over a ridge, and down into the next valley. Double-click to finish.
4. Compare the normal distance with the **Terrain (3D)** distance.
5. Now measure a path of the same length across the flat Ganga plains. Is the difference bigger or smaller?

Mountain path: flat _____ km, 3D _____ km Plains path: flat _____ km, 3D _____ km

FUN FACT

Walking up and down mountains really is longer than the map says. That's one reason a 20 km trek in the Himalayas can take a whole day!

7

Save Your Best View

Find your favourite mountain view and share it.

1. Pick a peak, valley, or river you like. Tilt and turn until the view looks amazing.
2. Try **Controls** → **Sun** and change the time of day. Watch the shadows stretch across the mountains at sunrise and sunset.
3. Open **Project** → **Print Layout...**, give your map a title, like "**Above the Himalayas**", and add a north arrow.
4. Export it as a **PNG** image.

TIP

A scale bar doesn't work well on a tilted 3D view, because things near you look bigger than things far away. That's why we only add a north arrow this time.

You Explored the Roof of the World!

You turned a flat map into a 3D landscape, measured real mountain heights, and saw why walking in the mountains takes so long.

What You Learned

Every place has an **elevation** above sea level

The shape of the land is called **terrain**, stored in a **DEM**

Looking at the map from the side in **perspective view**

Making shapes easier to see with **vertical exaggeration**

Reading heights from the map

Measuring distance along the ground in **3D**

Try It Yourself

- Follow the Ganga river upstream from the plains into the mountains. Can you find where it starts, near Gangotri?
- Fly down to the Western Ghats along India's west coast. How high are they compared with the Himalayas?
- Try **Controls** → **Flight Simulator** and fly through a mountain valley.
- Use **Record Map Tour** to make a short video flying over your favourite peaks.